AKENTEN APPIAH-MENKA UNIVERSITY
*of* Skills Training *and* Entrepreneurial Development

NIMDEE NE NSANONWUMA SUAKYERE MA MPUNTUO

# WELCOME BACK TO SCHOOL !!!...



## Course Information

- Course Code: ICT233
- Course Title: Web Technologies and Design
- Credit Hours: 3
- Lecturer: Maxwell Dorgbefu Jnr.

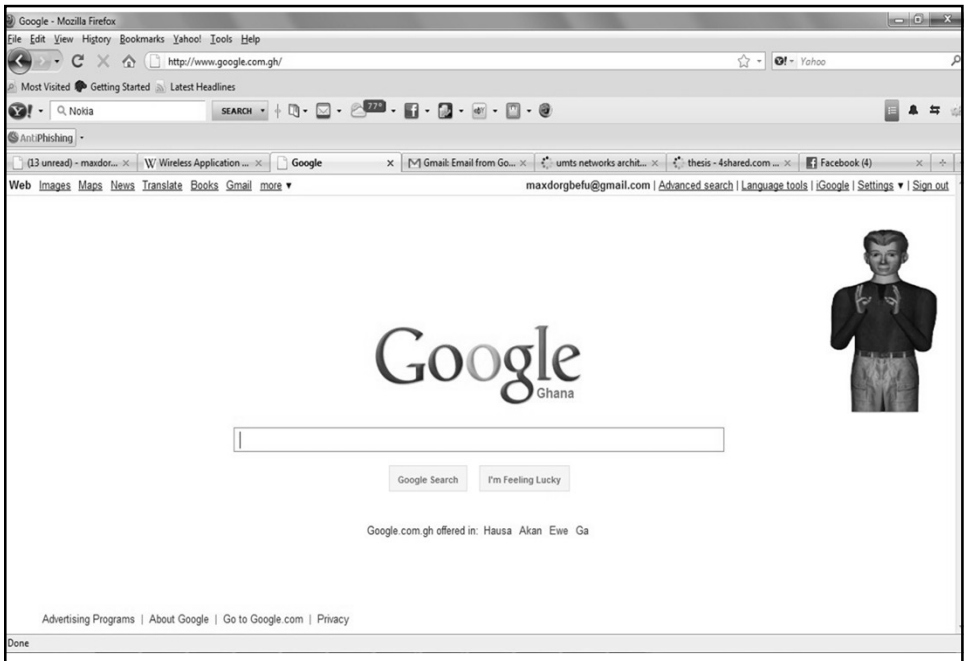
# ICT 233 - Thematic Areas

- Fundamentals of Web Technologies
- HyperText Markup Language (HTML)
- Cascading Style Sheets (CSS)
- JavaScript (JS)



# Today's Lecture

- Overview of Key Web Technologies
- The World Wide Web(WWW) and the Internet
- Domain Names and Uniform Resource Identifiers
- The WWW Model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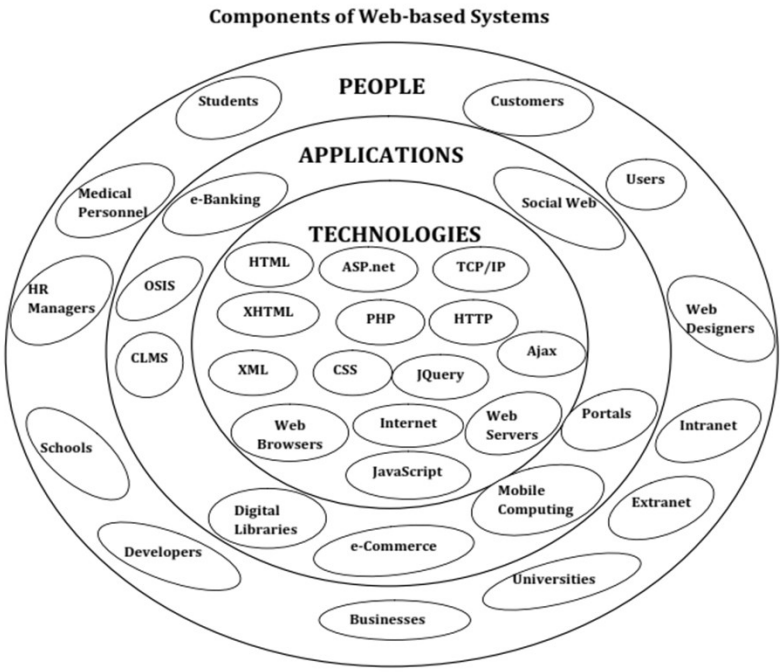

Components of Web-based Systems
PEOPLE
Students
Customers
APPLICATIONS
Users
Medical Personnel
e-Banking
Social Web
TECHNOLOGIES
HTML
ASP.net
TCP/IP
HR Managers
OSIS
XHTML
PHP
HTTP
Web Designers
CLMS
Ajax
XML
CSS
JQuery
Portals
Intranet
Web Browsers
Internet
Web Servers
Schools
JavaScript
Mobile Computing
Extranet
Digital Libraries
e-Commerce
Developers
Universities
Businesses

## What are Web Technologies?

- This is  a generic term used to describe a collection of tools that are used to plan, design, develop and deploy state-of –the-art websites and web applications.
- There are quiet a number of web technologies in use today by professional web developers.



## Some Popular Web Technologies

- The following web technologies are very relevant in the development of modern websites and web applications.
  - HTML
  - CSS
  - JavaScript
  - PHP
  - WordPress
  - Angular
  - Bootstrap
  - ReactJS
  - Laravel
  - MongoDB
  - Node.js

## SGML

- SGML stands for Standard Generalized Markup Language.
- SGML is a system(ISO standard) used to specify document structure.
- It was originally developed as part of the IBM document-sharing project in 1974 and became an official ISO standard in 1986.
- Most markup languages such as HTML and XML are derived from SGML.



## HTML

- HTML stands for **H**yper**T**ext **M**arkup **L**anguage.
- HTML is a language, which makes it possible to present information on the Internet.
- That is, web pages all over the world consist of HTML.
- However, HTML is not a programming language but a markup language.

## Sample HTML Code

```
<!DOCTYPE html>
    <html lang="en">
    <head>
        <title> My First HTML Page. </title>
        <meta name= "viewport" content= "width=device-
          width, initial-scale=1.0">
    </head>
    <body>
        <h1>Welcome to my first web page </h1>
        <p> I developed this web page using HTML, and this is the first
          paragraph of my web page. Please pay attention to the tags.
          You need a web browser such as Google Chrome to preview
          this page</p>
    </body>
</html>
```



## Cascading Style Sheets (CSS)

- CSS is a style sheets language (style language) introduced by W3C.
- Cascading refers to the process of determining the priority of styling rules.
- CSS is used to define the presentational semantics of structured documents.
- It provides control over visual as well as aural characteristics of HTML and XHTML documents and their elements.

# Sample CSS Code

```
body {
  background-color: lightblue;
}

h1 {
  color: white;
  text-align: center;
}

p {
  font-family: verdana;
  font-size: 20px;
}
```



# JavaScript(JS)

- JavaScript is an object-based client-side scripting language that adds interactivity and programmability to web pages.
- JS was created by Brendan Eich at Netscape.
- It was introduced in December 1995 and initially named "LiveScript".
- Licensing issues at the time between Netscape and Microsoft over "LiveScript" and "JScript", respectively, led to JavaScript's standardization as ECMAScript by Ecma International standards organization in June 1997.
- But JavaScript remains the preferred name of the language.

## Sample JavaScript Code

```
<!DOCTYPE html>
    <html lang="en">
    <head>
        <title> My First HTML Page. </title>
        <meta name= "viewport" content= "width=device-
          width, initial-scale=1.0">
    </head>
    <body>
        <h1>Welcome to my first web page </h1>
        <script>
          alert("Hello, this is a simple alert in JavaScript");
          console.log("This will be shown on the console");
        </script>
    </body>
</html>
```



## Extensible Markup Language (XML)

- Extensible Markup Language (XML) is a universal format for structured documents and data on the World Wide Web.
- XML, like HTML, is based on the granddaddy of all markup languages, Standard Generalized Markup Language (SGML).
- In contrast to most computer languages, XML has no fixed, predefined set of tags.
- With XML, individual markup formats can be defined.

## AJAX

- AJAX is an acronym for Asynchronous JavaScript and XML.
- It is not a programming language but a group of web development technologies related to each other, such as HTML, CSS, JavaScript, XML, and XSLT.
- Ajax can be used on the client side to create interactive web applications. Web site applications empowered with Ajax can send data to and retrieve data from servers asynchronously



## PHP

- PHP stands for **PHP: Hypertext Preprocessor** and is one of the most popular open-source server-side scripting languages.
- It is a cross-platform, general-purpose language originally created by **Rasmus Lerdorf** in 1994 for generating dynamic web pages.
- PHP is a widely used, free, and efficient alternative to competitors such as ASP.net.

## Sample PHP Code

```
<!DOCTYPE html>
    <html lang="en">
    <head>
        <title> My First HTML Page. </title>
        <meta name= "viewport" content= "width=device-
          width, initial-scale=1.0">
    </head>
    <body>
        <h1>Welcome to my first web page </h1>
        <?php
          echo "This is a PHP segment on an HTML page";
          echo "This file must be saved with the .php extension";
        ?>
    </body>
</html>
```



## Moment of Reflection

- Please reflect a bit and write down your answer to the question below:
- Question ??????
- What is the difference between the Internet and the web?

## What is the Web - 1

- The web is an information-sharing model involving a coordination of Internet servers that support some specially formatted documents.
- Documents are formatted using HTML standard.
- It uses HTTP as its communication protocol.
- Documents support links to other documents including graphics, audio and video files.



## What is the Web?

- The world wide web (web) is a network of information resources.
- The web relies on **three** mechanisms to make these resources readily available to the widest possible audience:
  1. A uniform naming scheme for locating resources on the web (e.g., URLs).
  2. Protocols, for access to named resources over the web (e.g., HTTP).
  3. Hypertext, for easy navigation among resources (e.g., HTML).

## History of the Web - 1

- The www was invented by **Tim Berners-Lee** and **Robert Cailliau**, two researchers working at CERN, the European Nuclear Research Centre (formerly, Centre European pour la Recherche Nucleaire), located in Geneva, Switzerland (1989).
- World Wide Web (WWW) project, started in 1990.



## History of the Web - 2

- **Tim Berners-Lee** and others created W3C as an industry consortium dedicated to building consensus around Web technologies
- Aimed at building a distributed hypermedia system embedded with resources, enabled for accessing on servers and built over the Internet.

## Domain Names

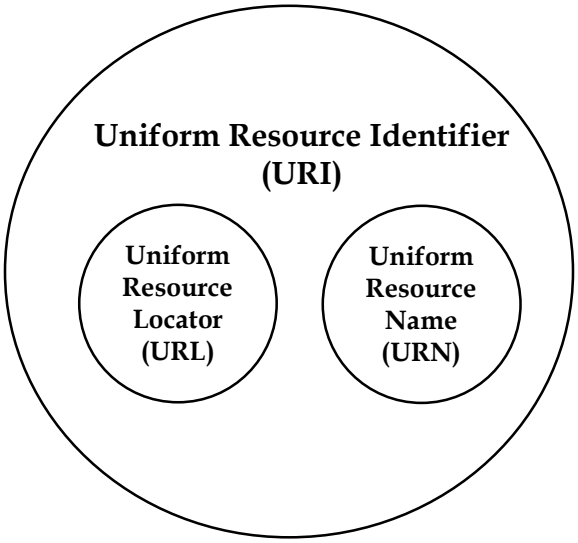
- Web resources can be located by unique IP addresses; however, IP addresses are difficult to remember.
- Consequently, domain names are used instead in most cases.
- Figure 4-1 shows the relationship between a domain name and a URL; ***www.example.com*** is a subdomain of the node ***example.com***, which is the subdomain of the ***com*** domain.
- The domain name syntax rules are defined by RFC 1035, RFC 1123 and RFC 2181.



## Uniform Resource Identifier (URI)

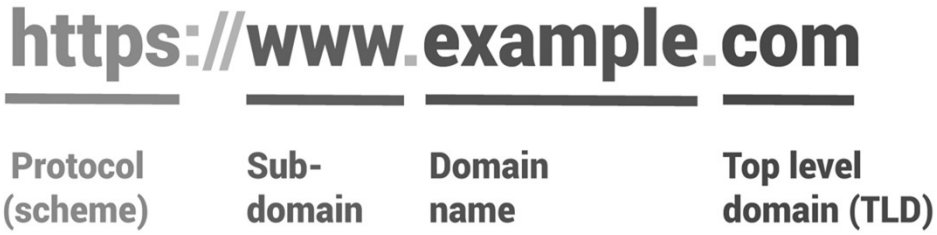
- A Uniform Resource Identifier (URI) is a character string that identifies a name or a resource on the Internet.
- URIs can be classified as Uniform Resource Locators (URLs) or Uniform Resource Names (URNs).
- URL provides a method or specifications for finding resources on the Internet (including protocol and path).
- URNs define a method for referencing an object without declaring the full path to the object.

# Uniform Resource Identifier (URI)






# Parts of URL

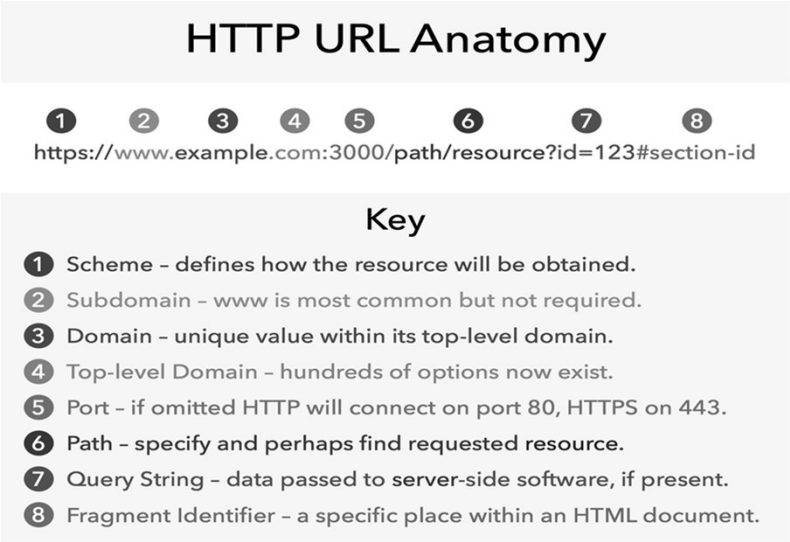


# Parts of URN

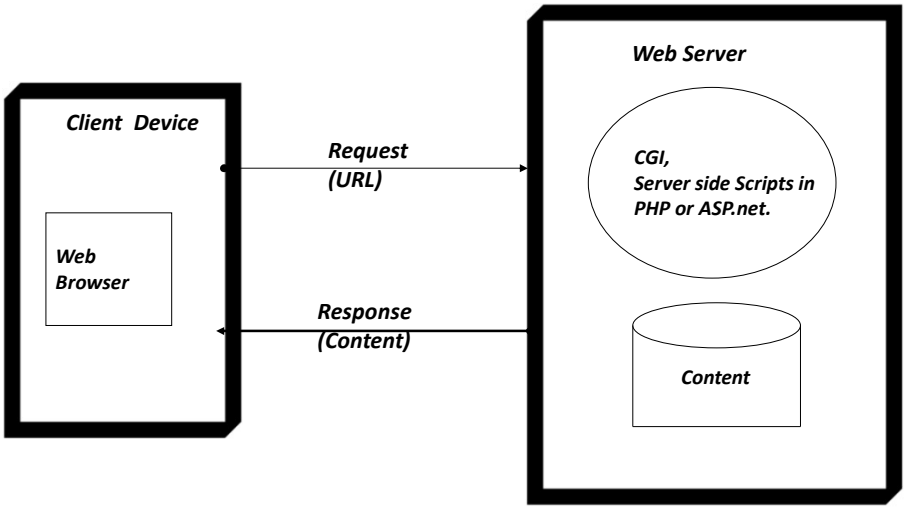
**Source**: *https://medium.com/@abhirup.acharya009/uri-vs-urn-vs-url-key-distinctions-explained-dec8e02ebd18*



# HTTP URL Details

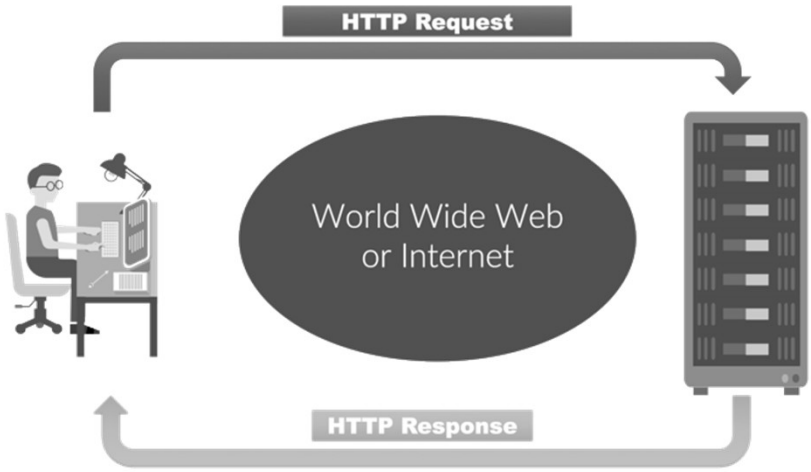


# The WWW Model






# HTTP Request and Response Model



# Focus Questions?

- Explain the following:
  - Web page
  - Index page
  - Web site
  - Web browser
  - Web application
- Briefly describe how you would register for a domain name.
- Write down any 10 web hosting companies you know of
- Provide the full meaning of the following acronyms as applied in web technologies
  - ICANN, JSON, AJAX, RSS, URI, XSLT



# END OF TODAY'S LECTURE

**PLEASE DO NOT MISS NEXT WEEK'S LECTURE ON:**

1. **Web Browsers and Servers**
2. **Domain Registration and Hosting**
3. **The Web Design Process**